

Pregnancy, cancer, and radiation—a modern refresher

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Abstract

Cancer occurs in ~1 per 1000 pregnancies; thousands of patients may require radiation procedures for diagnosis and treatment each year in the United States alone. This rare but high-risk scenario, coupled with fear of radiation, has created ambiguity in the ideal management of pregnant patients. Without a comprehensive guide for the use of radiation in imaging and treatment for pregnant cancer patients, it is difficult for providers to offer optimal patient-centered care without introducing disparities. The goal of this paper is to provide guidance on the use of radiation for screening, diagnosis, staging, and treatment of cancer, while highlighting gaps in existing knowledge and guidelines. The intention is that physicians, medical physicists, and patients could use this document as a resource for shared decision-making, ensuring safe and effective practice. A team of physicians and medical physicists with expertise in imaging, radiotherapy, and maternal fetal medicine was assembled, along with a patient advocate and lawyer. Existing guidelines and recent literature were reviewed. Authors also drew from their experience where published guidance was lacking. The resulting document discusses best practice to guide use of radiation for cancer, as well as patient-centered care management and legal considerations. Overall, it is possible to safely and effectively deliver radiation to pregnant patients in numerous circumstances. Use of radiation or other modalities should be discussed through shared decision-making with the physician and patient, contextualizing the maternal and fetal risk from treatments. Healthcare providers should support wide access to reproductive healthcare to allow equitable, evidence-based, patient-centered healthcare.

Introduction

Cancer occurs in approximately 1 per 1000 pregnancies.¹ This rate is rising, likely due to delayed childbearing and increased detection.^{1,2} Common cancers during pregnancy include melanoma, cancers of breast and cervix, lymphomas and leukemias.³ While pregnancy is not a risk factor for malignancy,^{4,5} diagnosis may be delayed if symptoms are falsely attributed to pregnancy.⁶

Managing cancer during pregnancy requires a multidisciplinary, patient-centered approach. The American Society of

Clinical Oncology (ASCO) provides a framework for navigating the medical, legal, and ethical challenges of cancer care during pregnancy,⁷ balancing optimal patient outcomes and fetal risk. ASCO recommends standard-of-care when possible and weighing risks to the patient and the fetus when considering deviation.

Oftentimes patients and physicians are uncomfortable with fetal exposure to radiation or chemotherapy. However, deviating from standard of care treatments can significantly impact maternal health. For example, treatment delays of 4 months or longer

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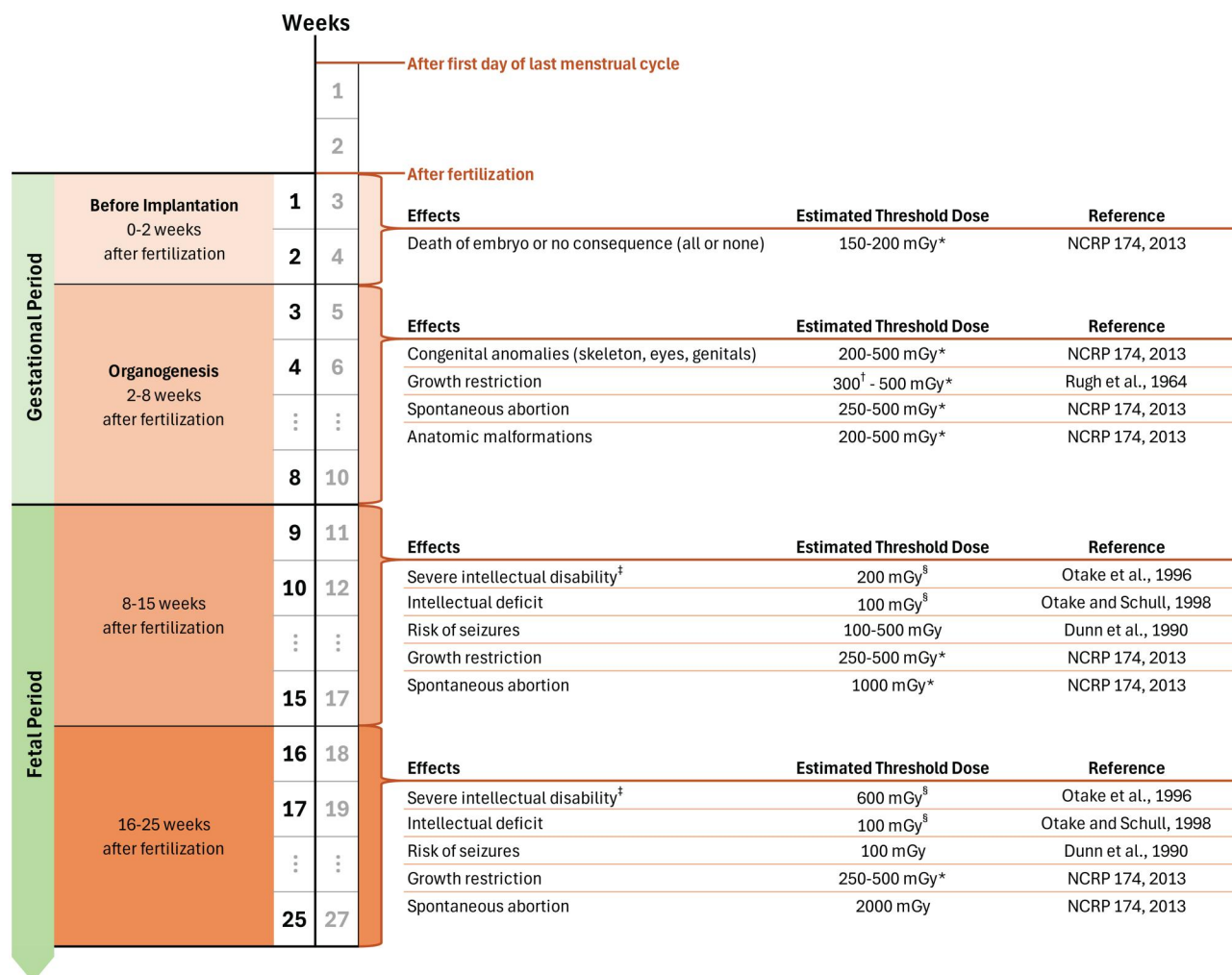
in cervical cancer are associated with a 2.31 times increased mortality risk.⁸ In most cases, pregnancy itself will not worsen prognosis unless treatment is declined by the patient due to fetal concerns. Unless indicated for the health of the pregnant patient, the American College of Obstetricians and Gynecologists and ASCO recommend avoiding delivery before 37 weeks.^{7,9} This highlights the critical importance of contextualizing fetal risk from cancer treatments, and the impact of any deviations from standard of care on both maternal and fetal health.

While there are ample resources for determining the role of systemic therapies, surgery, and other non-radiation treatments during pregnancy, the guidance for radiotherapy is sparse and scattered. Radiotherapy is a cornerstone of cancer treatment, and radiotherapy is beneficial in ~50% of cancer cases.¹⁰ In many cases, it can be delivered with very low doses to the fetus. The goal of this paper is to discuss how radiation can be safely considered for certain pregnant patients, including the vital importance of diagnostic imaging in appropriate staging, and to discuss the best practices to deliver treatment using modern approaches to minimize fetal risk. It concludes with discussions of patient-centered care and access issues

surrounding termination. We use terms like “pregnant individuals” and “patients” to acknowledge that not all pregnant people identify as women.

Radiation dose and fetal risk

Fetal risk depends on gestational age at time of exposure and radiation dose^{11,12} (Figure 1). Effects vary based on gestational age and stage of organogenesis at the time of exposure. Prior to 10 weeks from last menstrual period, spontaneous abortion, growth restriction, congenital anomalies, and anatomic malformations can occur at thresholds of 200 mGy or more. However after 10 weeks, the organs have been formed and are essentially immutable. Therefore, starting at 11 weeks, the fetal risks include seizure, growth restriction, spontaneous abortion, and intellectual deficits of varying degrees in accordance with dose, occurring at a range of dose thresholds from 100 to 2000 mGy. Deterministic radiation effects occur above a threshold dose, while stochastic effects may occur at any dose, and the probability of occurrence typically increases with dose.¹³ Stochastic effects include radiation-induced cancer, but the limited data



* Inferred from rat models.

[†] There is not enough information to definitively document effects in the human at doses to the embryo or fetus <0.3 Gy* (NCRP 174, 2013).

[‡] Defined as IQ < 70 or "unable to perform simple calculation, to make simple conversation, to care for himself or herself, or if he or she was completely unmanageable or had been institutionalized" (Otake et al., 1996).

[§] Risk of effect increases with dose.

Figure 1. Deterministic effects of ionizing radiation on the fetus by gestational age and estimated threshold dose.

suggest cancer risk from gestational exposure is significantly lower than exposure during early childhood.¹⁴ A 10-20 mGy fetal dose may increase the risk of leukemia by a factor of 1.5-2.0 over a background rate of ~1 in 3000.¹¹ Restated, there is a greater than 99% chance that a fetus exposed to 20 mGy will be unaffected,¹⁵ and a ~98% chance at 50 mGy. However, uncertainties in estimates below 100 mGy prevent accurate risk quantification.¹⁵

Pregnancy screening

Pregnancy screening for imaging is discussed in [Supplementary Materials S1](#). Screening is recommended prior to radiation therapy¹⁶ and each practice should develop a standardized policy. Screening should focus on the patient's physical and endocrinologic capacity for childbearing and participation in behaviors which can result in pregnancy. For external beam radiotherapy (EBRT), a negative pregnancy test should be secured within 14 days prior to treatment initiation.¹⁷ For therapeutic I-131 administration, a negative serum pregnancy test should be obtained 72 hours prior to treatment.^{12,18} Though urine pregnancy tests can be used, serum pregnancy tests can detect pregnancy earlier and with higher sensitivity.¹⁹ Knowledge of state regulations is helpful for pregnancy testing in minors, as the definition of minor and requirement for parental consent varies.¹²

It is important to counsel patients to prevent pregnancy during treatment and after radiopharmaceutical administration.^{12,18-20} Some radiopharmaceuticals are excreted in breastmilk and, generally, breastfeeding should cease for all I-131 labeled radiopharmaceuticals, Ga-67 citrate, and therapeutic administrations. Recommendations regarding the duration of interruption/cessation of breastfeeding following radiopharmaceuticals vary, and an Authorized User/Nuclear Medicine Physician should be consulted.²⁰⁻²²

Recommendations and best practices for imaging

Imaging is critical in cancer screening, diagnosis, staging, and monitoring of treatment response. Clinicians should carefully consider the use of imaging modalities that may impact the patient's course of treatment, in the context of fetal risk from ionizing radiation. Dose optimization in imaging pregnant patients should include the same approaches used in non-pregnant patients: proper x-ray field collimation and patient positioning, limiting the number of views and phases to those necessary, and only performing exams that will impact the course of care.^{15,23-25} [Figure 2](#) describes imaging protocol modifications for specific modalities.

Radiation doses

[Table 1](#) summarizes typical fetal doses for exams associated with cancer screening, diagnosis, and staging,^{32,42,43} though the exact dose depends on imaging parameters, pregnancy stage, and patient habitus. Any x-ray-based exam where the gravid uterus is not in the field-of-view (FOV) results in a fetal dose below 1 mGy; no imaging protocol modifications are necessary. For most x-ray-based exams where the fetus is in the FOV, fetal dose is under 10 mGy.¹⁵ Exceptions include computed tomography (CT) and fluoroscopy-guided interventions (FGIs), where doses can exceed 50 mGy.¹⁵ If multiple imaging exams are necessary, cumulative dose should be considered.

Given that fetal doses are typically well below 100 mGy, deterministic radiation effects are not relevant^{12,15,16,26,32} and stochastic risk is negligible or non-existent. External shielding of the fetus for x-ray-based imaging is no longer recommended as it minimally reduces dose but may interfere with dose optimization tools or degrade image quality.⁴⁴

Benefits and risks of imaging modalities

Considerations relevant to cancer care are discussed below, with more detail in [Supplementary Materials S2](#).

Mammography

Pregnant patients should follow breast cancer screening guidelines for the general population.^{11,12,45} While physiologic changes of pregnancy and lactation may degrade the diagnostic efficacy of mammography, this does not warrant a screening delay.⁴⁵ Screening mammography is essential in detecting subclinical breast cancers during pregnancy.⁴⁶

Ultrasound

Ultrasound has 100% sensitivity in patients with a palpable breast mass. Specificity may be reduced, as breast cancer in pregnant patients is more likely to mimic benign lesions on ultrasound than in non-pregnant patients.⁴⁷ Any solid mass identified on ultrasound should proceed to biopsy.⁴⁸

Contrast-enhanced ultrasound (CEUS) is useful for assessing tumors and lymph nodes with metastatic infiltration⁴⁹ and for imaging patients with cardiotoxic chemotherapy regimens. Limited studies have shown that microbubbles in ultrasound contrast agents do not cross the placental barrier, and no adverse reactions have been observed.⁵⁰ Despite no known risks, the American College of Radiology (ACR) only recommends CEUS use when there is a clear patient benefit.⁵¹

MRI

Non-contrast MRI is considered safe during pregnancy when the static magnetic field is 3 T or less and the scanner is in normal operating mode (whole body-averaged specific absorption ratio [SAR] limited to 2 W/kg).⁵² Despite limited evidence of fetal risk from gadolinium-based contrast agents (GBCA),⁵³ current recommendations discourage their use except in situations where the anticipated clinical benefit justifies the unknown risk of fetal exposure to GBCA.^{11,54}

X-ray

Planar radiography can be safely performed without modifications, given that fetal doses are low ([Table 1](#)).

Fluoroscopy and FGIs

Diagnostic fluoroscopy may facilitate cancer diagnosis or treatment through biopsy or port placement for chemotherapy²³ and is used to treat or manage conditions that are common in pregnancy, such as iliofemoral deep vein thrombosis and embolism.²³ When the fetus is outside the FOV, FGI fetal dose will be below 1 mGy. However, if the fetus is in the FOV during a prolonged procedure, fetal dose may exceed 100 mGy. The benefits and risks should be weighed against alternative treatments. Fetal time in the FOV should be limited.

Computed tomography

No protocol adjustments are required when the fetus is outside the FOV as fetal doses are below 1 mGy. For multi-phase exams where the fetus is inside the FOV, modifications should be

IMAGING PROTOCOL MODIFICATIONS FOR PREGNANT PATIENTS		
No protocol modifications are needed, regardless of the anatomy being imaged		
MAMMOGRAPHY		
	No protocol modifications needed	
PLANAR X-RAY		
	No protocol modifications needed	
Protocol modifications depend on whether the fetus is in the imaging field of view		
	If the fetus is INSIDE the field of view	If the fetus is OUTSIDE the field of view
CT	Only include necessary phases	No protocol modifications needed
		
FLUOROSCOPY	Minimize the time the fetus is in the field of view	No protocol modifications needed
		
Protocol modifications should be considered for ALL pregnant patients, regardless of the anatomy being imaged		
MRI	Use a field strength of 3T or below Maintain whole body-averaged SAR below 2 W/kg Avoid gadolinium contrast agents	
		
NUCLEAR MEDICINE IMAGING*	I-131 is contraindicated in pregnant patients Consider decreasing the injected activity while increasing the scan time to maintain image quality	
		
*The information in this figure only applies to diagnostic nuclear medicine exams. Radiopharmaceutical therapy is generally contraindicated in pregnant patients (see Brachytherapy, radiopharmaceutical, and internal therapy).		

Figure 2. Summary of imaging protocol modifications for pregnant patients.

considered, eg, reducing the scan range or conducting a single-phase study.

Iodinated contrast agents

Iodinated contrast crosses the placenta but does not affect short-term neonatal thyroid stimulating hormone (TSH). Long-term effects on TSH function are unknown, but iodinated contrast agents are recommended during pregnancy when needed for accurate diagnosis.⁵⁴ Importantly, management of contrast reactions may be different for pregnant patients.^{54,55}

Nuclear medicine—diagnostic

Fetal doses in nuclear medicine depend on radiopharmaceutical accumulation in the fetus and nearby patient organs.⁵⁶ Positron emission tomography (PET) is the most common diagnostic

nuclear medicine exam in this population as cancers that are common during pregnancy exhibit strong F-18-FDG uptake.⁵⁶ Most diagnostic nuclear medicine studies use low administered activities of radiopharmaceuticals with short half-lives, resulting in fetal doses below 5 mGy.¹² Fetal thyroid doses can be substantial for I-131 (over 40 Gy), which is contraindicated for diagnosis and therapy during pregnancy.^{11,12}

Recommendations and best practices for radiation therapy
Introduction

Many cancers outside the pelvis may be safely treated with radiation (Table S1), while pelvic radiation may cause severe or lethal fetal consequences.^{16,57} While no strict dose limitation guidelines

Table 1. Estimated fetal doses from imaging exams performed during cancer staging, diagnosis, and follow-up.

Examination	Fetus in field of view (FOV)?	Estimated fetal/uterine dose (mGy) ^a	References
Radiography			
Chest (1-view)	No	0.001	26,27
Abdomen, pelvis, or lumbar spine (1-view)	Yes	0.1-3	26-28
Diagnostic Fluoroscopy & FGIs			
Diagnostic barium enema study	Yes	1.0-20	26,28
IVC filter placement	Yes	2-4	29
Percutaneous nephrostomy	Yes	7	30
Mammography			
Screening or diagnostic mammography	No	<0.01	28,31
Stereotactic breast biopsy	No	<0.01	28,31
CT			
Head or neck	No	0-0.005	26,32
Chest	No	0.01-0.82	26,28,32-36
Abdomen	Yes	1.3-35	26,28,32
Colonography	Yes	9-16	37,38
Abdomen/pelvis	Yes	10-50	26,28,32
Nuclear Medicine and PET	Administered Activity		
Tc-99m sulfur colloid lymphoscintigraphy	18.5 MBq	<0.01-0.05	39,40
	92.5 MBq	0.03-4.3	
I-123 NaI scintigraphy	7.4 MBq	0.01-0.6	26,41
I-131 thyroid uptake	0.55 MBq	0.03-0.15 (whole body)	16
		275-600 (thyroid)	
I-123 thyroid uptake	30 MBq	0.3-0.6 (whole body)	16
		150-450 (thyroid)	
Tc-99m DMSA scintigraphy	222 MBq	0.75-1.12	24,41
I-131 metastases imaging	40 MBq	2-11 (whole body)	16
		20 000-44 000 (thyroid)	
F-18 FDG PET only	370 MBq	3-10	24,41
Ga-67 citrate scintigraphy	185 MBq	18-38	41

^a Cited studies may report fetal or uterine dose, estimated using phantoms or computational models. To estimate an overall risk, given the numerous uncertainties, it can be assumed that the values shown represent whole-body fetal dose unless otherwise indicated.

exist, most sources claim deterministic effects may occur above 100 mGy.^{16,58} Therapeutic radiation should aim to keep fetal dose below this threshold. For extremity, head and neck, and brain sites fetal doses can be kept below 100 mGy with few additional accommodations, but fetal dose should still be closely monitored.¹⁴ For chest, abdomen, and upper thigh sites, fetal doses may exceed 100 mGy, necessitating shielding and other fetal-dose reduction strategies (see below).¹⁵ When fetal dose is likely to exceed 100 mGy, patient risks from foregoing radiation should be weighed against fetal risks.

Several relevant guidelines from professional organizations and consensus groups exist^{7,59-65}; however, many contain limited radiation-specific information^{7,62-64} or are outdated.⁶⁰ The American Association of Physicists in Medicine (AAPM) Task Group Report 36 contains recommendations for estimating and reducing fetal dose,⁶⁰ but does not discuss modern treatment modalities.⁶⁶ The French Society for Radiation Oncology practice guidelines on radiotherapy and pregnancy⁵⁹ reported that supradiaphragmatic radiation can be safely delivered, and phantom and in vivo dose measurements should be performed. Multiple review papers have also been published,^{57,67-70} but many contain limited⁶⁷⁻⁶⁹ and/or misleading radiation information⁷⁰ (eg, avoiding radiation except in “rare scenarios in which the benefits exceed the risk”⁶⁷). In many cases the benefit of radiotherapy outweighs the fetal risks. Recommendations specific to common disease sites are described below.

Disease site considerations

Head and neck

Radiotherapy is often important for primary tumor and nodal irradiation. Some sites may require radiotherapy alone or with chemotherapy due to severe toxicities of surgery. Given the

distance between the treatment site and the uterus, fetal dose can be kept below 100 mGy, and treatment decisions should focus on optimizing patient outcome.⁷¹⁻⁷³

Central nervous system

Radiotherapy during pregnancy is most relevant for high grade primary central nervous system (CNS) tumors and brain metastases.⁷⁴ For brain metastases, the primary cancer type, extra- and intracranial extent and location, and systemic therapy options need to be closely weighed, as there are few options that completely obviate the need for radiotherapy. Many aggressive primary brain neoplasms can be surgically resected, but adjuvant radiotherapy improves survival.⁷⁴ If diagnosed early in pregnancy, radiotherapy should not be delayed in aggressive histologies as recurrence occurs quickly.⁷⁵ Fortunately, the large distance from the uterus typically results in low fetal doses.^{71,73}

Thoracic

Most early stage cancers can be managed with surgical resection and/or chemotherapy, and local recurrence risks without radiotherapy are typically low.⁷⁶ However, more extensive lung cancers are managed with primary radiotherapy, which can potentially be safely delivered with proper shielding and dosimetric fetal dose reduction strategies.^{73,77} Treatment planning should consider pregnancy-related changes in pulmonary function.⁷⁸

Breast

Diagnosis of breast cancer during pregnancy is often delayed and presents at a more advanced stage than in nonpregnant patients.^{79,80} A full staging workup is essential as advanced disease may impact treatment decisions. Treatment management should follow standardized protocols for non-pregnant patients

as closely as possible,⁷ and typically includes surgery followed by radiation and systemic therapy. Radiotherapy decisions should consider gestational age, as fetal exposure increases later in pregnancy as the uterus becomes closer to the treatment field, balanced against the impact of delaying radiation, which may impact local control. There is no consensus of acceptable surgery-to-radiotherapy interval, with evidence for permissible delays varying from 6 to 20 weeks.^{81,82} Often, the standard treatment course may involve neoadjuvant chemotherapy, which may result in delaying radiotherapy till after pregnancy. If prompt radiotherapy is indicated, breast radiotherapy can be delivered relatively safely during the first and much of the second trimesters,^{61,65,83} and there is ample evidence showing fetal doses are often below 100 mGy using shielding.^{77,84-87}

Lymphoma

Treatment options vary depending on the cancer type and stage, gestational age, and maternal symptoms, but generally treatment occurs in the second and third trimester.^{62,88,89} Chemotherapy is the most common treatment, but supradiaphragmatic lymphomas can be safely treated with radiation if necessary.^{90,91} In some cases of asymptomatic Hodgkin lymphoma, treatment can be deferred until after delivery.⁹²

Cervical cancer

The timing and manner of treatment is based on gestational age, tumor stage, and maternal wishes.^{63,93} For stages 1B1 or less, options include delay or more limited surgical approaches.⁹³ For stages greater than 1B1, delivering standard-of-care chemoradiotherapy would induce abortion in 1-2 months.⁹⁴ A short treatment delay to allow lung maturation could be considered in some third trimester cases.⁹⁵ Local abortion restrictions and fetal personhood laws should be considered to avoid prosecution of the treatment team or the patient. If laws or patient wishes prevent termination, pelvic lymphadenectomy or neoadjuvant

systemic therapy could be considered, risking inferior maternal outcomes.⁹⁶

External beam radiotherapy of pregnant patients

An alternative to photon EBRT is pencil beam scanning proton therapy (PBS-PRT). PBS-PRT can produce fetal equivalent doses over ten times lower than photon EBRT for brain and head and neck cancers.⁹⁷⁻¹⁰² When PBS-PRT is not available, it is possible to deliver photons or electrons using 3D conformal radiotherapy (CRT), intensity modulated radiation therapy (IMRT), volumetric modulated arc therapy (VMAT), CyberKnife, GammaKnife, and linac-based stereotactic radiosurgery (SRS), while maintaining fetal doses below 100 mGy (Table S1).^{72,103-106} Though reports during pregnancy are limited, electrons have been used successfully for intraoperative breast cancer treatment¹⁰⁷ and chest wall irradiation.¹⁰⁸ Passive scattering proton therapy (PS-PRT) produces doses ten times greater than photon therapy and is not recommended during pregnancy.⁹⁷

Radiation doses from EBRT

Fetal dose from photon EBRT sites outside the uterus is due to internal patient scatter, treatment head leakage and collimator scatter (Figure 3).¹⁰⁹ Leakage and collimator scatter can be minimized through shielding and monitor unit (MU) reduction, whereas internal scatter can only be decreased by increasing the distance to the fetus or decreasing field size.¹⁰⁹ Beyond the field edge, out-of field dose decreases approximately exponentially.^{109,110}

For electron beams, out-of field dose is mainly caused by scattered electrons originating from the primary beam or trimmer and collimator scatter.¹⁰⁹ Electron beam out-of-field dose, particularly for certain linear accelerators, can exceed that of photons. For photons and electrons, out-of-field dose is highest superficially, then decreases rapidly with depth, remaining fairly constant at deeper depths.^{109,112} Secondary neutron dose for electron beams is typically ~5% of that from similar photon energies.¹⁰⁹

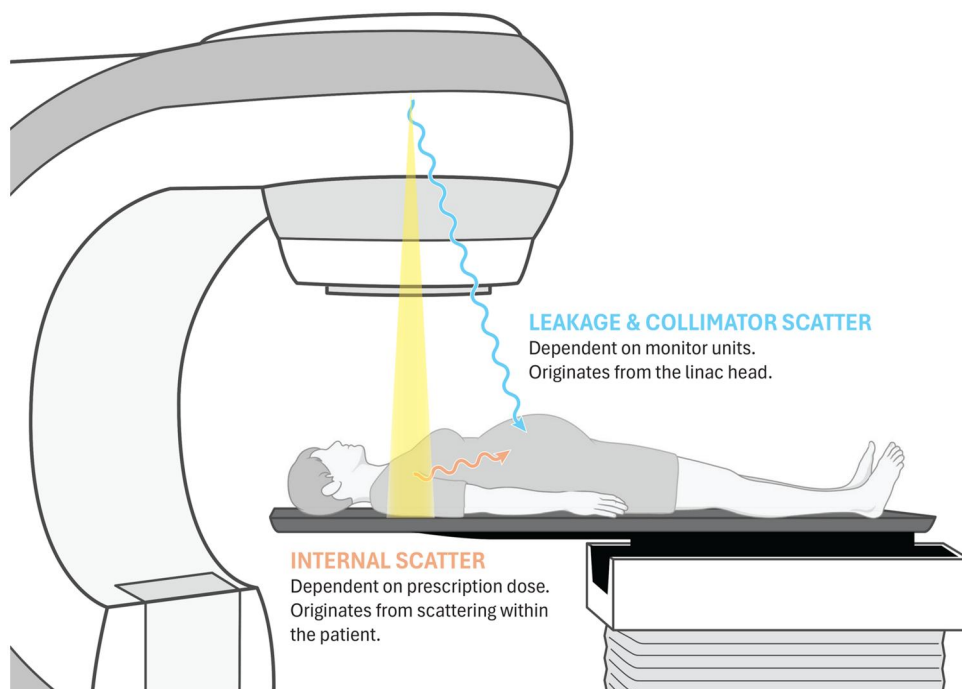


Figure 3. Sources of fetal dose from photon EBRT. Patient graphic created in BioRender. Juang, T. (2025) <https://BioRender.com/30ghnfu>

In PRT, fetal dose primarily results from internal neutrons and secondary photons produced from proton interactions in the patient. Neutrons comprise the majority of out-of-field dose with photons contributing $\sim 10\%$.¹⁰⁹

Treatment planning

Photons

Physicists, dosimetrists and physicians should collaborate to determine the best plan for a given patient and treatment site. 3DCRT features lower out-of-field dose than IMRT/VMAT due to fewer MUs.^{101,111} In sites that necessitate tight conformality, eg, head and neck or brain, IMRT/VMAT may be used with MU reduction efforts.¹⁰⁶ Flattening filter free beams are preferred due to reduced flattening filter scatter and reduced target current, decreasing fetal dose by $\sim 50\%$.^{111,113} Beam energies of 10 MV or less are preferred to eliminate neutron dose.⁶⁰

Protons

In general, beam angles can be selected to generate a robust plan based on setup and range uncertainties. If this can be achieved without a vertex beam then fetal dose can be further minimized.^{98,99} Use of beam modifying devices will increase fetal dose (eg, by 30% with range shifters⁹⁹).

Electrons

Fetal dose can be minimized by selecting a low energy beam and avoiding directing the gantry towards the fetus.¹⁰⁹

Dose estimation and measurements

Physicists should determine and document the potential fetal dose from simulation through treatment, including prior diagnostic imaging dose. When possible, the fetus should be excluded from simulation CT FOV. Computational estimates, though useful for assessing treatment options, should not replace phantom measurements in determining patient-specific fetal dose.⁶⁶ Most commercial treatment planning systems report inaccurate doses 3 cm beyond the field edge¹¹⁴ and should not be used to calculate fetal dose.¹⁰⁹ Monte Carlo simulations with full scatter conditions have higher accuracy,¹¹⁵ but require proprietary treatment machine information that is often not readily available.¹¹⁶ When referring to literature values, it is important to account for differences between linear accelerator models.⁶⁶

Treatment plans under consideration should be delivered on a phantom, with setup imaging, to estimate total fetal dose and guide plan selection. In vivo dosimeters should be placed on the surface under bolus to compare to in vivo dosimetry. TG158 discusses detector selection for out-of-field dosimetry.¹⁰⁹ Published phantom setups include Rando torso phantoms next to solid water stacks for ion chamber measurements (Figure 4)^{59,117} and 3D printed phantoms.¹¹⁸ It may be necessary to deliver multiple fractions to reach the lower limit of detection of the detectors.⁶⁰ Measurements should be performed with and without shielding being considered.⁵⁹

For photons and electrons, ion chambers should be placed beyond the depth of maximum dose¹¹¹ at 3 distances from treatment isocenter to simulate the locations of the fundus, umbilicus, and pubic bone.⁶⁰ Fetal dose measurements for PRT should include contributions from neutrons and photons. A common detector is the wide-energy neutron detection instrument (WENDI-II), which can measure both neutrons and photons and reports dose equivalent.¹¹⁹ Neutron dose estimates feature greater uncertainty due to biological effectiveness conversions and less accurate detector calibrations than ion chambers.^{120,121}



Figure 4. Example phantom fetal dose measurement setups used at Mayo Clinic. (A) Photon EBRT setup for a brain plan using an anthropomorphic phantom, with an ion chamber at 5 cm depth in solid water behind a lead shield. (B) PBS-PRT setup using an anthropomorphic phantom, WENDI-II neutron meter placed at the phantom's abdomen, and slices of acrylic. Part B by Dupere, et al is licensed under CC BY-NC-ND 4.0.¹⁰⁰

Shielding

Shielding with the equivalent of 5 half value layers (HVLs) of lead can reduce fetal dose up to 50% for photon EBRT, and nearly eliminate setup imaging dose.¹¹¹ Shielding only reduces head leakage and collimator scatter; therefore, it is most effective for high MU modalities, eg, IMRT/VMAT.¹⁰⁴ Photon shielding setups can be 3D printed or crafted from equipment readily available in the clinic (see Figure 5 and^{104,111,122,123} for examples). For electrons, the scattered electrons primarily causing out-of-field dose can be shielded by adding water equivalent bolus of thickness E (MeV)/4 (cm).¹¹²

Lead shielding is ineffective for PBS-PRT.^{109,124} Additionally, neutron shielding has not been used during PBS-PRT because the majority of fetal dose arises from internal patient scatter,^{98,99,101,124} and constructing shields of appropriate thickness is challenging.

Treatment

Fetal dose should be monitored during treatment,^{59,60} with dosimeters placed under bolus on the patient's skin at the fundus, umbilicus, and pubic symphysis.⁶⁰ TLDs and OSLDs are well-suited for photon and electron beams.¹⁰⁹ Diodes and MOSFETs may be used with consideration for the low energy and low dose



Figure 5. Mobile shield used at MD Anderson for linac-based treatments of pregnant patients.

nature of out-of-field radiation.¹⁰⁹ Film is not recommended due to limited sensitivity below 1 mGy.¹⁰⁹ For protons, OSLDs such as Landauer Luxel+ with a Neutrak CR-39 plastic nuclear track detectors can measure both neutrons and photons; however, they must be sent out for readout. A discrepancy between phantom and in vivo measurements should be investigated, especially as doses approach 100 mGy.

Image guidance

Image guidance is a vital component of modern radiotherapy. Setup imaging fetal dose over treatment courses is low, ranging from 0.08 mGy for daily kV setup imaging to 64 mGy for daily kV and simulation CT.^{99,102,125} A single high-dose cone-beam CT (CBCT) can deliver 0.125, 0.05, and 0.02 mGy at 2, 7, and 12 cm outside of the imaging field, respectively.¹²⁶ Uterine doses from single, standard head and chest CBCTs are 0.2 and 0.6 mGy, and 0.1 and 0.2 mGy for low-dose techniques.¹²⁷ To our knowledge, there are no publications reporting fluoroscopy or 4D-CBCT for EBRT image guidance during pregnancy. Imaging doses may vary significantly with technique, geometry, and fetal distance. Imaging should be selected based on patient-specific needs, using the lowest-dose modality adequate for setup and monitoring.¹²⁷

Brachytherapy, radiopharmaceutical, and internal therapy

Reports or guidelines for brachytherapy during pregnancy are sparse, but data are promising for non-pelvic sites. Pregnant patients with uveal melanoma were successfully treated with low dose-rate (LDR) plaque brachytherapy, with fetal doses under 0.002 mGy.¹²⁸⁻¹³⁰ High dose-rate (HDR) Ir-192 for accelerated breast brachytherapy resulted in an estimated 20 mGy fetal dose, when the breast was placed inside a lead box to reduce direct scatter.⁸⁴

Shielded brachytherapy could reduce fetal dose compared to EBRT for targets greater than 12 cm from the fundus.⁸⁴ Prior dose calculations at large distances (0-60 cm) from various brachytherapy sources could be applied to estimate fetal dose.¹³¹ During HDR treatments, the fetus should be shielded from transit doses.

In general, most therapeutic radiopharmaceutical (RP) interventions, including Zevalin, Xofigo, I-131, and Lutathera, are contraindicated during pregnancy.⁴¹ Selective Internal Radiation Therapy (SIRT) including Sir-spheres and Theraspheres (Y-90) is also contraindicated; however, a patient treated at 17 weeks' gestation with microspheres for fibrolamellar hepatocellular carcinoma delivered a healthy baby, though no dose estimate was given.¹³²

Considerations for care delivery

Patient-centered care

A cancer diagnosis is emotionally challenging, especially for pregnant patients with concerns about impacts on the fetus.¹³³ Providers must offer evidence-based guidance, discussing data limitations, and support patient decision-making around imaging, treatment, early delivery, or pregnancy termination. Emphasizing that maternal health is essential for fetal health, standard-of-care should be prioritized⁷ by the treatment team. Patients may decide on a non-standard of care pathway for themselves, as always, for any reason. It is critical that these decisions are made with the appropriate risk assessment of the treatments available. Using the techniques and data assembled in this manuscript, patients will find that the risks with radiation are manageable for many pregnant individuals.

Legal considerations regarding abortion

Pregnancy termination may be crucial medical care for cancer patients and is chosen by 9-28% of pregnant cancer patients.¹³⁴ In 2024, the ACR passed a resolution affirming the need for abortion access "as an evidence-based medical care option," consistent with professional practice guidelines.¹³⁵ Inability to access abortion may complicate treatment. Since the U.S. Supreme Court's *Dobbs v. Jackson Women's Health Organization* (2022) decision, many US states have enacted laws restricting abortion access, including total or near-total bans. Penalties for violation can include criminal charges, fines, license suspension, or sentences up to life in prison.¹³⁶ While some laws include exceptions for threats to the pregnant person's health, these are often vague and narrow.^{137,138} This leads to confusion and fear among patients and physicians,¹³⁹ and reluctance of physicians to offer needed care (so called "hesitant medicine,"¹⁴⁰) as they would bear the burden of proof to justify the abortion.

In this shifting legal landscape, clinicians should protect themselves while providing evidence-based care. They should understand their employer's stance on this issue and review their liability insurance to ensure they do not violate their coverage policies. If time allows, they should engage their risk management and ethics departments on choosing the medically appropriate path forward that is also legally defensible and appropriate. Ideally, hospitals in affected regions will develop a rapid and efficient process to support physicians in making evidence-based, patient centered care decisions throughout the spectrum of care options. Accurate, thorough documentation is essential, with particular care to consult notes and informed consent. Clinicians should stay informed and seek guidance from organizations offering legal guidance related to abortion care.

Looking forward

As cancer management grows more complex, so do considerations for pregnant patients. Updated guidance is needed for advanced radiation modalities (eg, PRT, brachytherapy), with a framework to address new technology and modalities. Multidisciplinary teams with expertise in treatment during pregnancy are essential to avoid disparities in care, including medical physicists. Resources like the Advisory Board on Cancer, Infertility and Pregnancy offer free multidisciplinary expert consultations (<https://www.ab-cip.org/>).

It is now more important than ever to ensure physicians are properly informed regarding fetal risk from radiation, balanced with the patient risk from withholding necessary imaging and treatment. With clinical trials unlikely, registries, such as The International Network on Cancer, Infertility, and Pregnancy,¹⁴¹ should be utilized to track outcomes and improve understanding of fetal radiation risks and impacts of treatment modifications on pregnant patients.

Conclusions

Radiation is a critical component in the screening, diagnosis, staging, and treatment of cancer. Despite potential exposure to the fetus, it is often possible to safely and effectively use radiation with pregnant patients. While general guidance is provided, the use of radiation or other care options should be determined on a case-by-case basis in shared decision making with the patient. Future efforts should consider the impact of legal reproductive restrictions on delivering equitable, evidence-based, and patient-centered healthcare.

Author contributions

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Supplementary material

Supplementary material is available at JNCI Cancer Spectrum online.

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Conflicts of interest

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Data availability

No new data were generated or analyzed for this Commentary.

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